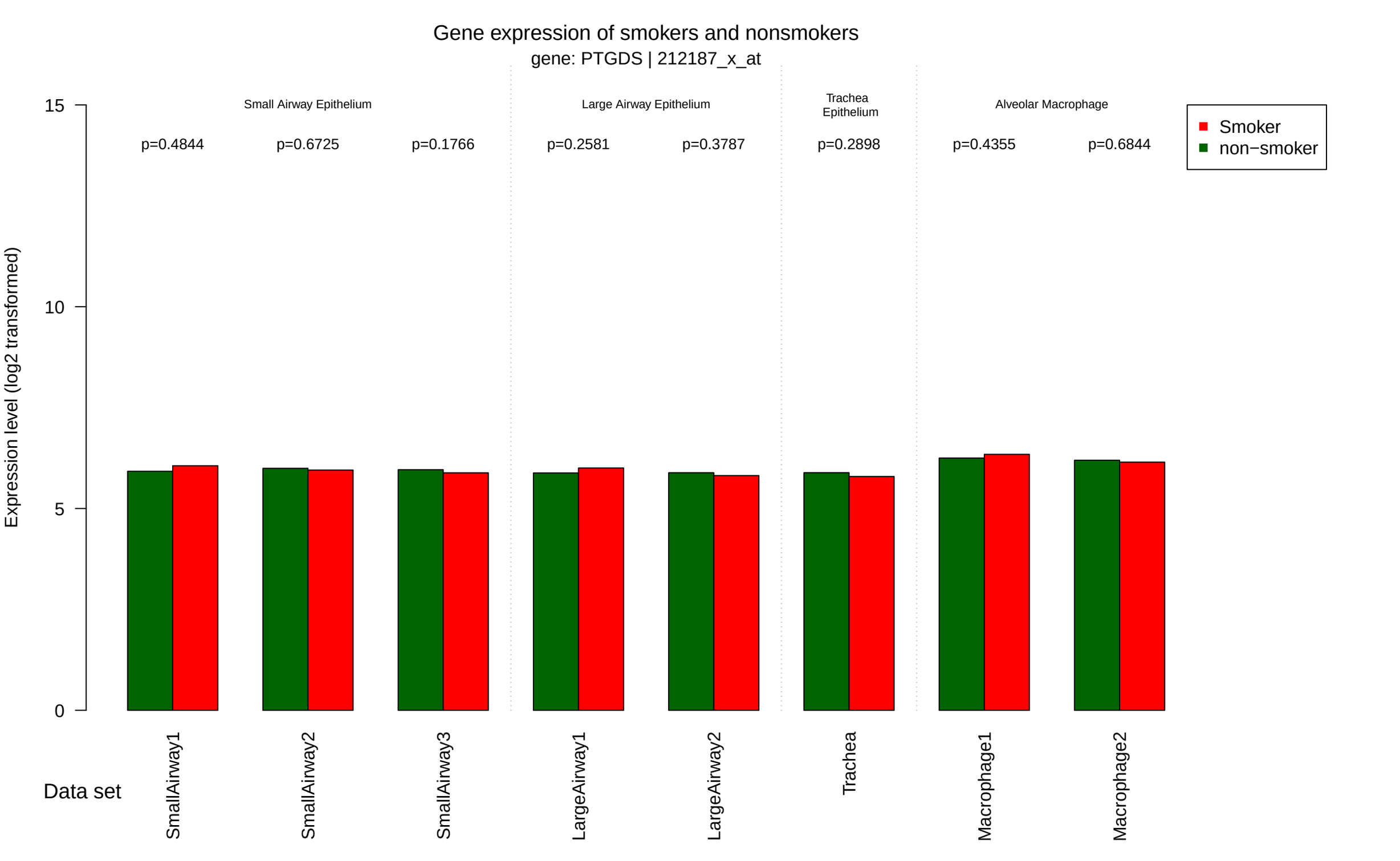
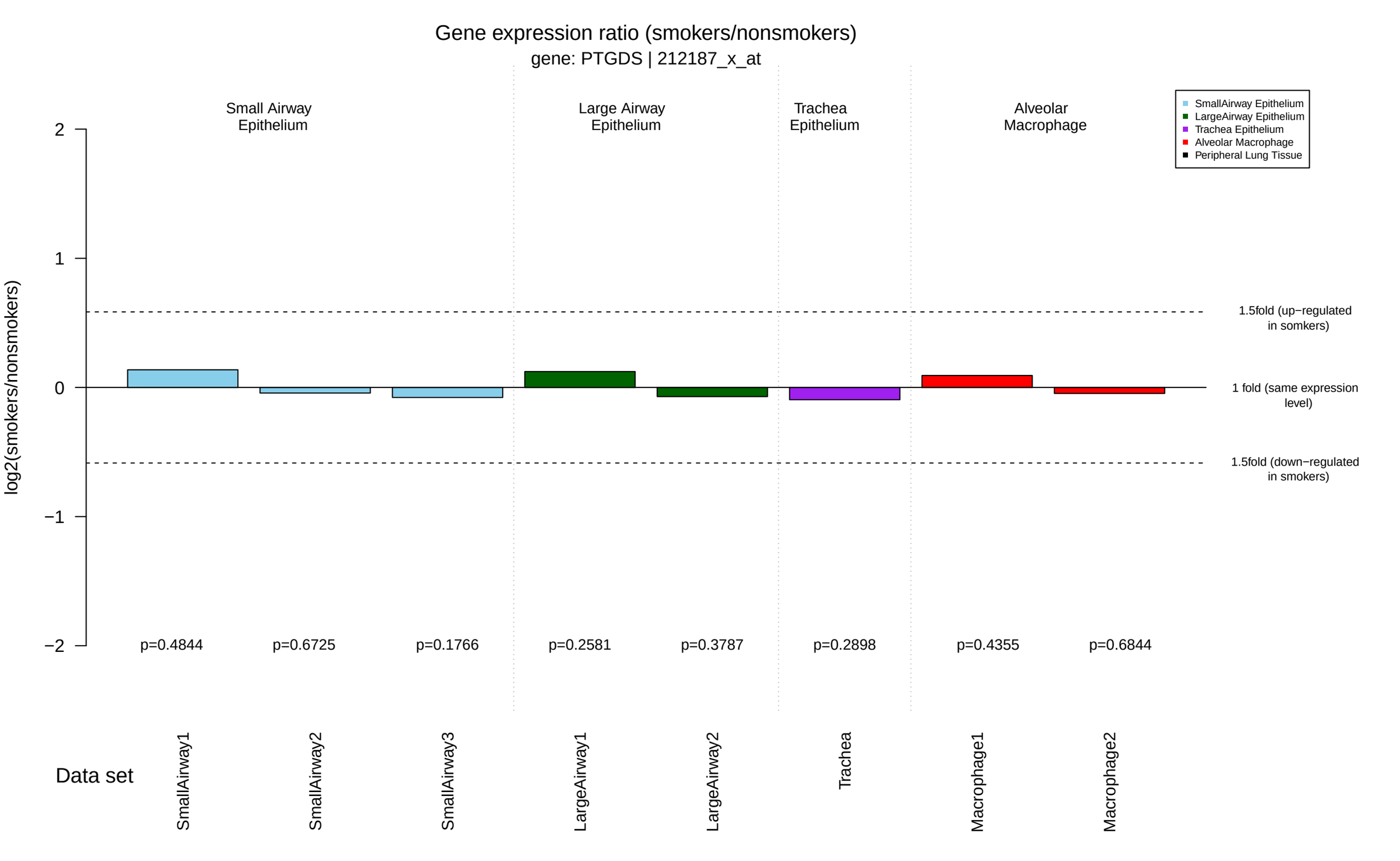
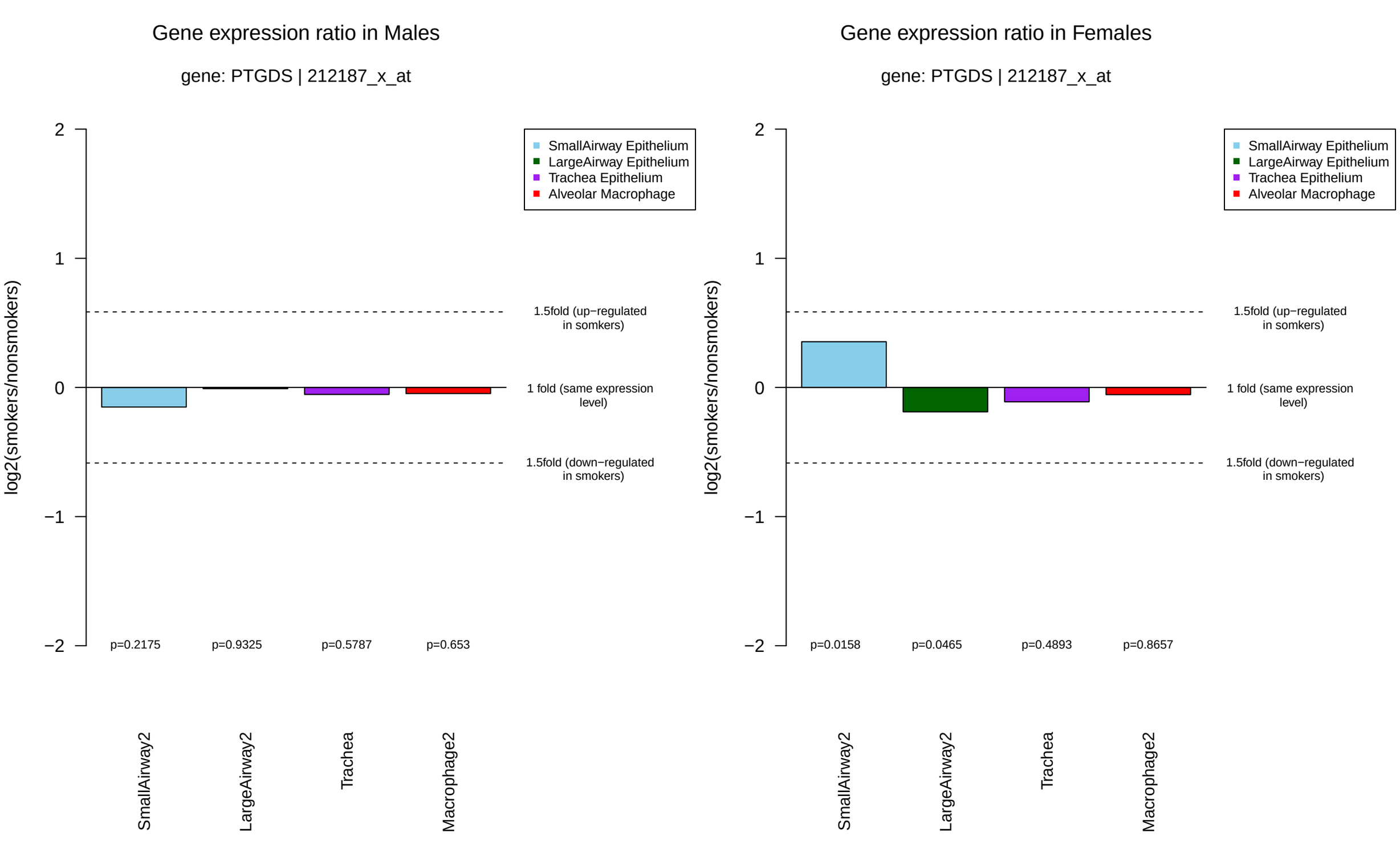
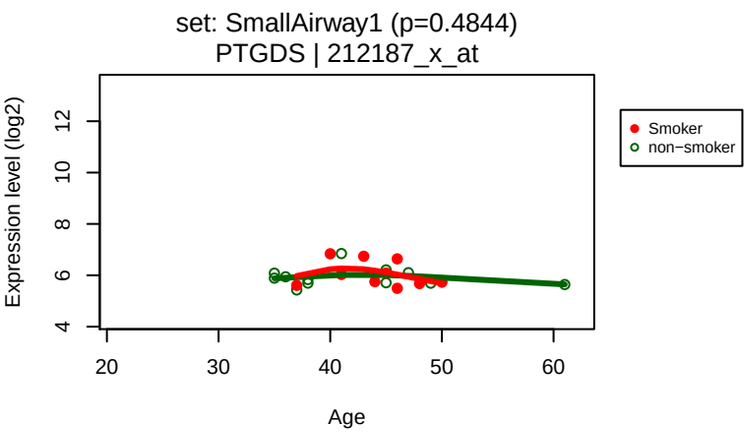
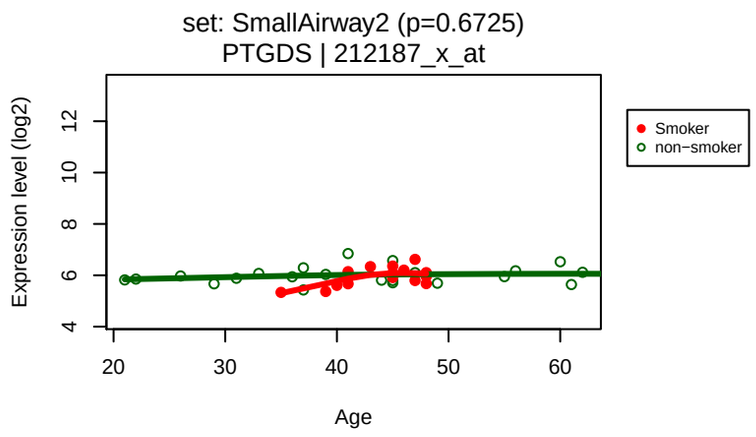
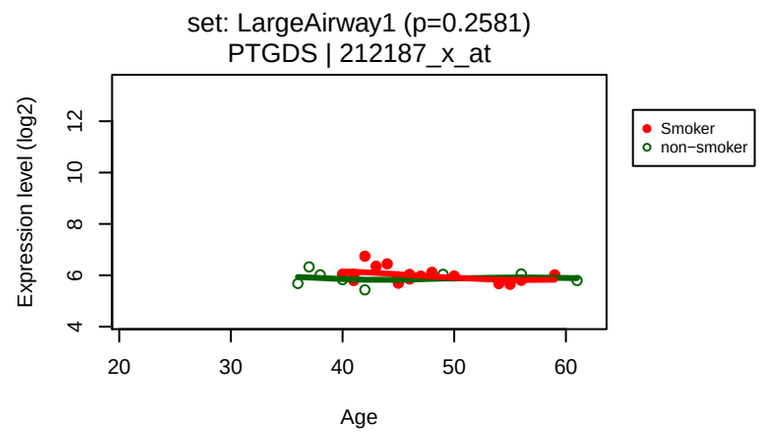

Gene expression of smokers and nonsmokers
gene: PTGDS | 212187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4844
p=0.6725
p=0.1766
p=0.2581
p=0.3787
p=0.2898
p=0.4355
p=0.6844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

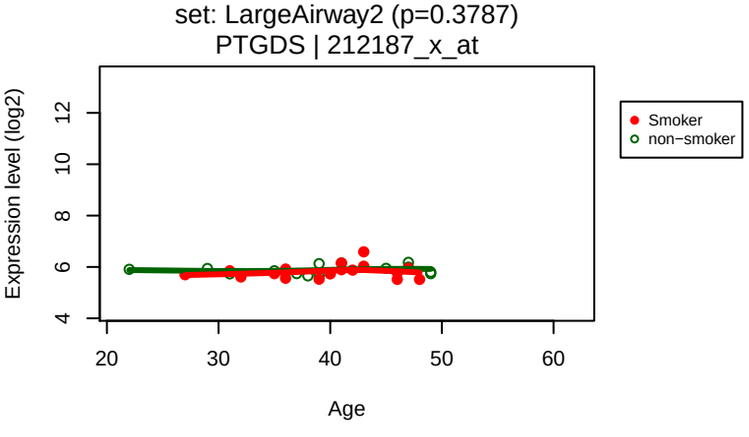

Gene expression ratio (smokers/nonsmokers)
gene: PTGDS | 212187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4844
p=0.6725
p=0.1766
p=0.2581
p=0.3787
p=0.2898
p=0.4355
p=0.6844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

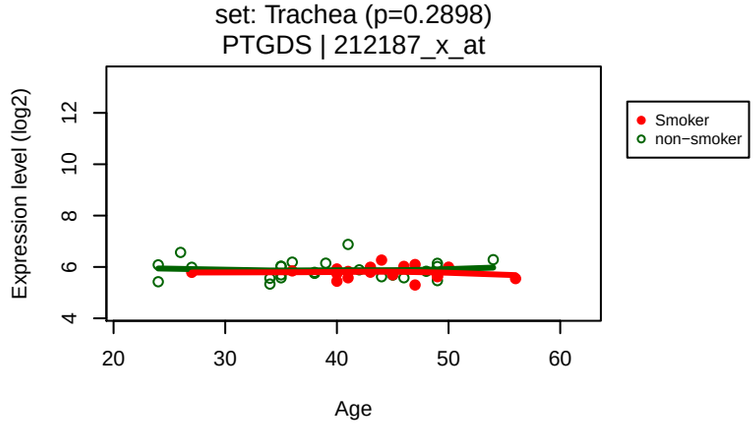

Gene expression ratio in Males
gene: PTGDS | 212187_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2175
p=0.9325
p=0.5787
p=0.653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGDS | 212187_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0158
p=0.0465
p=0.4893
p=0.8657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

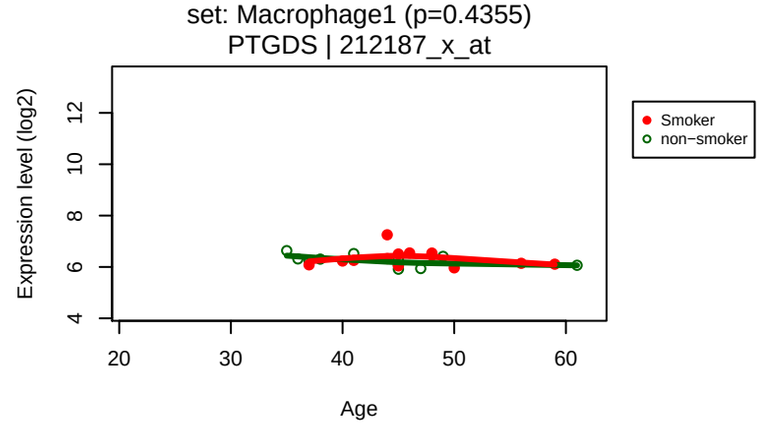

set: SmallAirway1 (p=0.4844)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker

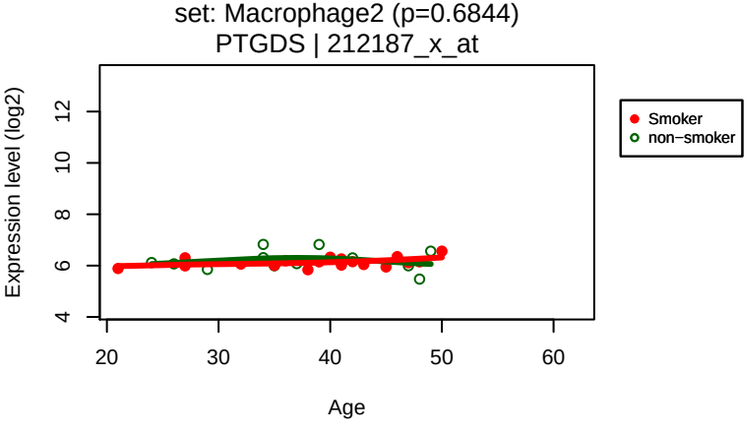

set: SmallAirway2 (p=0.6725)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker

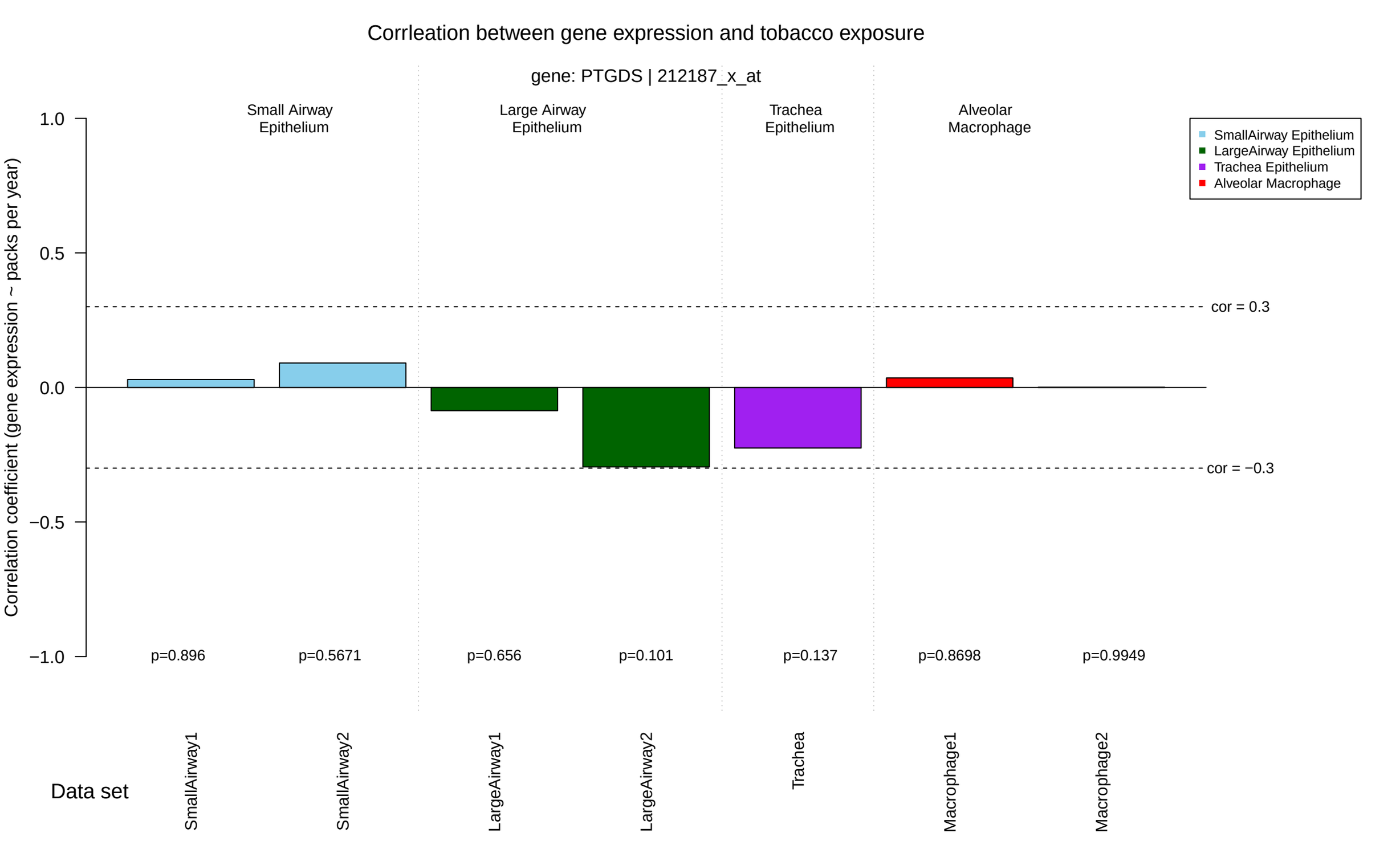

set: LargeAirway1 (p=0.2581)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3787)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2898)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4355)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6844)
PTGDS | 212187_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGDS | 212187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.896
p=0.5671
p=0.656
p=0.101
p=0.137
p=0.8698
p=0.9949
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2